

Que 1. Find the inverse of matrix by employing elementary transformation

$$(i) A = \begin{bmatrix} 2 & 3 & 4 \\ 4 & 3 & 1 \\ 1 & 2 & 4 \end{bmatrix} \text{ [2020-21]} \quad (ii) B = \begin{bmatrix} 3 & -3 & 4 \\ 2 & -3 & 4 \\ 0 & -1 & 1 \end{bmatrix} \text{ [2018-19]} \quad (iii) C = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 2 & 3 \\ 3 & 1 & 1 \end{bmatrix} \text{ [2017-18]}$$

Que 2. Reduce matrix A into normal form and hence find its rank

$$(i) A = \begin{bmatrix} 1 & 3 & 4 & 2 \\ 2 & -1 & 3 & 2 \\ 3 & -5 & 2 & 2 \\ 6 & -3 & 8 & 6 \end{bmatrix} \text{ [2019-20]} \quad (ii) B = \begin{bmatrix} 1 & 2 & -1 & 4 \\ 2 & 4 & 3 & 4 \\ 1 & 2 & 3 & 4 \\ 1 & -2 & 6 & 7 \end{bmatrix} \text{ [2018-19]}$$

Que 3. Find the value of b for which the rank of matrix $A = \begin{bmatrix} 2 & 4 & 2 \\ 3 & 1 & 2 \\ 1 & 0 & b \end{bmatrix}$ is 2. [2019-20]

Que 4. Find the rank of the matrix $A = \begin{bmatrix} 2 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{bmatrix}$ [2018-19]

Que 5. Reduce the matrix $\begin{bmatrix} 1 & 1 & 1 \\ 3 & 1 & 1 \end{bmatrix}$ into normal form and find its rank. [2021-22] , [2017-18]

Que 6. Find the rank of matrix by employing elementary transformation $A = \begin{bmatrix} 2 & -1 & 3 & -1 \\ 1 & 2 & -3 & -1 \\ 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & -1 \end{bmatrix}$ [2017-18]

Que 7. Test the consistency of the following system of equations and if system is consistent, solve them
 $x + y + z = 6$, $x + 2y + 3z = 14$, $x + 4y + 7z = 30$ [2022-23]

Que 8. For what values of μ and ω the simultaneous equations $x + y + z = 6$, $x + 2y + 3z = 10$, $x + 2y + \omega z = \mu$ have (a) No solution (b) Unique solution (c) Infinite many solutions.

[2019-20] [2017-18] [2015-16]

Que 9. Find the value of k for which the system of equations $(3k - 8)x + 3y + 3z = 0$, $3x + (3k - 8)y + 3z = 0$,

$3x + 3y + (3k - 8)z = 0$ has a non-trivial solution. [2021-22]

Que 10. For what values of λ and μ the system of linear equations $x + y + z = 6$, $x + 2y + 5z = 10$, $2x + 3y + \lambda z = \mu$ have (a) No solution (b) Unique solution (c) Infinite many solutions. Also find the solution when $\lambda = 2$ & $\mu = 10$. [2021-22]

Que 11. Find the non-singular matrix P and Q s.t. PAQ is in normal form where $A = \begin{bmatrix} 1 & 1 & 2 \\ 1 & 2 & 3 \\ 0 & 1 & 1 \end{bmatrix}$ [2020-21]

Answers

1. (i) $A^{-1} = \frac{1}{5} \begin{bmatrix} -10 & 4 & 9 \\ 15 & -4 & -14 \\ -5 & 1 & 6 \end{bmatrix}$ (ii) $B^{-1} = \begin{bmatrix} 1 & -1 & 0 \\ -2 & 3 & -4 \\ -2 & 3 & -3 \end{bmatrix}$ (iii) $C^{-1} = \frac{1}{2} \begin{bmatrix} 1 & -1 & 1 \\ -8 & 6 & -2 \\ 5 & -3 & 1 \end{bmatrix}$

2. (i) Rank A=3 (ii) Rank B=4

3. $b = \frac{3}{5}$

4. Rank A=1

5. Rank A=2

6. Rank A=4

7. $x = k - 2$, $y = 8 - 2k$, $z = k$, where k is arbitrary

8. (i) No solution $\omega = 3$, $\mu \neq 10$ (ii) Unique solution $\omega \neq 3$, μ may have any value

(iii) Infinite many solution $\omega = 3$, $\mu = 10$

9. $k = \frac{2}{3}, \frac{11}{3}, \frac{11}{3}$

10. (i) No solution $\lambda = 6$, $\mu \neq 16$ (ii) Unique solution $\lambda \neq 6$, μ may have any value

(iii) Infinite many solution $\lambda = 6$, $\mu = 16$, $x = \frac{13}{2}$, $y = -2$, $z = \frac{3}{2}$

11. $P = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 1 & -1 & 1 \end{bmatrix}$ and $Q = \begin{bmatrix} 1 & -1 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix}$